Nutrient solution dosing unit Specification

Document number	SHV-AG-SPEC-0102
Revision	B
Status	APPROVED
Classification	INTERNAL
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Approved by	Rosalind Ferrante
Date	2023-06-21

1. Purpose

This specification defines the performance, interface and verification requirements for the nutrient solution dosing unit (AG-NSD) of the Hydroponics & Food Production on Starhaven Orbital Habitat & Shipyard.

2. Scope

Fabrication and installation are performed under work package WP-SHV-AG-03 (WBS 1.11.3), owned by Treva Zander.

The prime supplier for flight hardware is Chiewab Medical. Responsible engineer: Carys Renner.

3. Applicable documents

Document	Title
SHV-SYS-SPEC-0001	Starhaven Master System Specification
SHV-AG-DWG-0202	Nutrient solution dosing unit general arrangement drawing
SHV-AG-BOM-0302	Nutrient solution dosing unit bill of materials
SHV-QA-PLN-0004	Starhaven Verification and Test Plan
SHV-QA-STD-0002	Station Workmanship Standard
SHV-ICD-PD-AG	Interface Control Document: Power Distribution / Hydroponics & Food Production
SHV-ICD-LS-AG	Interface Control Document: Life Support & Atmospherics / Hydroponics & Food Production
SHV-ICD-HA-AG	Interface Control Document: Habitation Blocks / Hydroponics & Food Production

4. Description

Hydroponics and food production grow fresh produce for the resident crew on stacked grow tiers under controlled lighting, and prepare meals in the central kitchens.

The nutrient solution dosing unit is delivered as a set of line-replaceable assemblies, integrated on station by Construction Operations and verified by Quality Assurance & Test.

5. Performance requirements

ID	Requirement	Value	Tolerance	Verification
AG-NSD-R01	The nutrient solution dosing unit shall provide a dosing accuracy of 2.4 %.	2.4 %	±2%	Inspection

ID	Requirement	Value	Tolerance	Verification
AG-NSD-R02	The nutrient solution dosing unit shall provide a dosing lines of 29 ea.	29 ea	±0.5%	Demonstration

5.1 General requirements

ID	Requirement	Verification
AG-NSD-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.	Inspection
AG-NSD-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.	Analysis
AG-NSD-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.	Inspection
AG-NSD-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.	Analysis
AG-NSD-G05	Documentation: Acceptance data packages shall be delivered with every unit.	Inspection

6. Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7. Verification

Each requirement is verified by the method stated in section 5, following SHV-QA-PLN-0004. Acceptance test reports are filed under the AG test-report series.

8. Revision history

Rev	Date	Description	Author
A	2023-02-14	Initial release.	Carys Renner
B	2023-06-21	Updated requirement values after design review.	Carys Renner